**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 121 · Lesson 121 · Writing Algorithms with Validation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why must a trace table record changes rather than only final output?

Q2. After pushing A then B onto a stack, which is popped first?

Q3. Why must a binary search operate on ordered data?

Q4. An OR gate has inputs 0 and 1. What is the output?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

validation — Checking that input meets rules.

robustness — Ability to handle invalid/unexpected input safely.

iteration — Repetition based on loop condition.

boundary — Input at/near permitted limit.

## New learning

Define constraints — State valid range, type and required behaviour before coding.

Build control flow — Use sequence, selection and iteration with clear termination.

Prove with tests — Use normal, boundary, erroneous and extreme cases appropriately.

## Worked example

Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.

1. Model answer — age = int(input("Age")) — while age < 11 OR age > 19 — endwhile — print(age) — Note: this handles numeric range, not a non-numeric input error without extra handling. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

Develop a loop that accepts a score from 0 to 100 inclusive, rejecting out-of-range numbers.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

score = int(input("Score"))

while score < 0 OR score > 100

endwhile

print(score)

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Explain why OR, not AND, joins the invalid conditions.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Write an algorithm accepting a positive integer (at least 1) and outputting it. [6]

### Question 2

Explain why 'WHILE age < 11 AND age > 19' fails as range validation. [6]

### Question 3

Extend a validation algorithm to reject non-integer input, describe robust behaviour and supply three tests. [8]

## Exit Ticket · Answer before checking

When rejecting values outside a range, the two invalid inequalities should usually be joined with AND.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It reveals execution path and when variable values change.

Q2. B, because a stack is LIFO.

Q3. It discards half the search space using an ordering comparison.

Q4. 1.

### You Do pages · Independent questions

Q1. Read integer [2], loop while <1 with re-prompt [2], terminate and output [2].

Q2. Cannot be below 11 and above 19 simultaneously [3]; use OR to detect either invalid condition [3].

Q3. Type checking/error handling [3], range validation [2], three labelled normal/boundary/erroneous tests [3].

### Exit Ticket · Model answer

False. An invalid input is below the minimum OR above the maximum.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**